Uniform and Exponential Distributions

CSU, Chico Math 350

2026-04-29

outline

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Uniform Distribution (Ch. 4.5.1)

Exponential Distribution (Ch 4.5.2)

References

Situation

Uniform random variables can be either discrete or continuous, and describe a distribution where all outcomes are equally likely. Examples include:

- the result of a die roll
- pseudo random number generation
- round off error in measurements

Random variable: Let X be a random variable from a Uniform distribution defined by the parameters a for the lower bound, and b for the upper bound.

Uniform Distribution Properties

Distributional Notation
$X \sim Unif(a, b)$

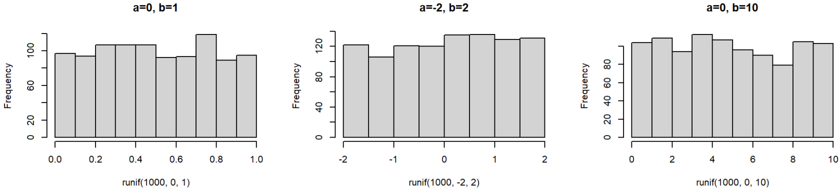
PDF
$f(x) = \frac{1}{b-a} \quad a \leq x \leq b$

Mean & Variance
$E(X) = \frac{b+a}{2}$
$Var(X) = \frac{(b-a)^2}{12}$

R Commands:
`punif(x, a, b)` to compute $P(X \leq x)$ (the CDF)
`runif(N, a, b)` to randomly draw N samples from a $X \sim Unif(a, b)$ distribution.

Visualizing the Shape

Notice how the distribution changes as the parameters do.



Example 1

Consider the PDF $f(x) = 1/11$, for $-3 < x < 8$ (and 0 for all other x

1) Calculate $P(0 \leq X \leq 4)$. Do this by hand, and confirm your answer using `punif()`.

By hand:

$$\int_0^4 \frac{1}{11} dx = \frac{1}{11} x \Big|_0^4 = \frac{4}{11}$$

```
punif(4, -3, 8) - punif(0, -3, 8) # 4/11
## [1] 0.3636364
```

Example 1

2) Calculate the mean and standard deviation. Do this by hand, and confirm your answer using simulation.

$$E(X) = \frac{b+a}{2} = \frac{8-3}{2} = 2.5$$

$$SD(X) = \sqrt{\frac{(b-a)^2}{12}} = \sqrt{\frac{(8+3)^2}{12}} = 3.17$$

```
x <- runif(10000, -3, 8)
mean(x)
## [1] 2.495369
sd(x)
## [1] 3.190014
```

Exercise 1

Suppose that a random variable X has a uniform distribution on the interval [-4,10]. Write down the PDF of X, then find the mean, standard deviation, and the value of $P(-1 < X < 6)$ both theoretically and using simulation.

outline

Situation

Exponential random variables measure the waiting time until the first event occurs in a Poisson process. Examples include:

- the waiting time until an electronic component fails
- the time between customers in a store

Random variable: Let X be a random variable from an Exponential distribution defined by the parameter λ for the rate.

Exponential Distribution Properties

Distributional Notation
$X \sim Exp(\lambda)$

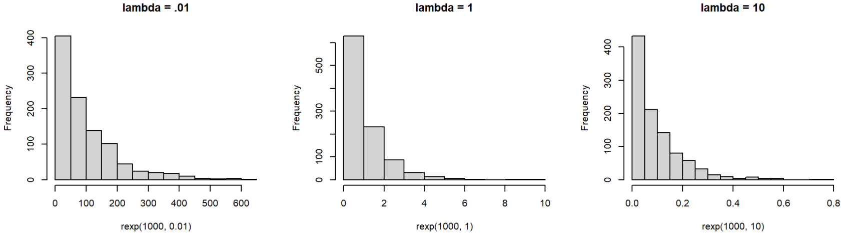
PDF
$f(x) = \lambda e^{-\lambda x} \quad x \geq 0$

Mean & Variance
$E(X) = \frac{1}{\lambda}$
$Var(X) = \frac{1}{\lambda^2}$

R Commands:
`dexp(x, lambda)` to compute $P(X == x)$
`pexp(x, lambda)` to compute $P(X \leq x)$ (the CDF)
`rexp(N, lambda)` to randomly draw N samples from a $X \sim Exp(\lambda)$ distribution.

Visualizing the Shape

Notice how the distribution changes as the parameters do.



Example 2

Suppose the time to failure (in years) for a particular component is distributed as an exponential random variable with rate $\lambda = 1/5$.

For better performance, the system has two components installed, and the system will work as long as either components installed, and the system will work as long as either component is functional.

Assume the time to failure for the two components is independent. What is the probability that the system will fail before 10 years has passed?

First, set up the problem by defining the random variables involved:

Let X_1 be the time until component 1 fails.
Let X_2 be the time until component 2 fails.

$$X_1 \sim Exp(1/5), \quad X_2 \sim Exp(1/5)$$

Example 2

The system fails if $X_1 < 10$ **and** $X_2 < 10$. Since these are *independent* events:

$$P(systemfailure) = P(X_1 < 10) * P(X_2 < 10)$$

Theoretical:

$$\int_0^{10} \frac{1}{5} e^{-\frac{1}{5}x_1} dx_1 * \int_0^{10} \frac{1}{5} e^{-\frac{1}{5}x_2} dx_2$$

Good news: `pexp` can handle the integration here:

```
pexp(10, 1/5)^2
## [1] 0.7476451
```

Simulation:

```
time.of.failure <- rexp(10000, 1/5)
mean(time.of.failure<10)^2
## [1] 0.7511689
```

Exercise 2: Bus Ride

Buses arrive at a particular stop at a rate of 4 per hour. Let X be the length of time in minutes that commuters have to wait at the bus stop until they see their bus arrive. *Hint: Be mindful of units!*

1) Calculate the mean and standard deviation of the wait time in minutes using both theoretical formulas and simulation.

2) Find the probability that a commuter waits less than 8 minutes for their bus to arrive.

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References

references I

Speegle & Clair. Probability, Statistics, and Data: A Fresh Approach Using R, 2021.

Also adapted from notes by Dr. Roualdes, Dr. Donatello, and Dr. Gray.